

Installation and data acquisition

ImSpector deconvolution mode

To utilize the deconvolution algorithm within MACS® iQ View – 3D Large Volume, it is necessary to capture your data using the **deconvolution mode in ImSpector**. The following steps provide guidance on activating the deconvolution mode, followed by instructions on how to acquire data within this mode.

Requirements

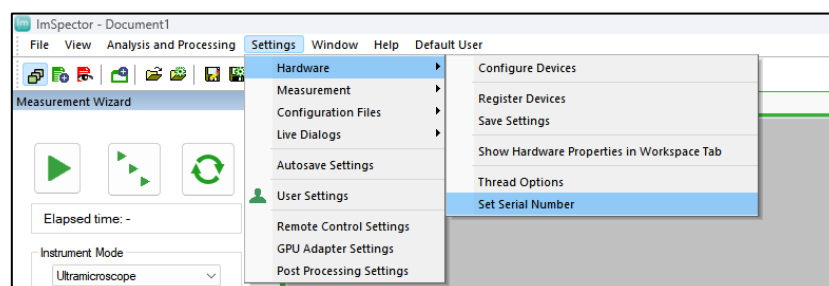
- ImSpector 7.6
- An instrument specific calibration file (*umb_light_sheet_calibration.ini*)
 - If the Ultramicroscope Blaze's serial number is **UM-3163 or higher**, the calibration file is already installed on your acquisition computer. Consequently, you can **proceed directly to step 3** and activate the deconvolution mode.
 - If the Ultramicroscope Blaze's serial number falls **below UM-3163**, our Technical Support team will provide you with the system-specific calibration file that you have to install on your acquisition computer as described in **step 1**.

Setup & Installation

Please download the latest version of ImSpector (ImSpector 7.6) on the UltraMicroscope Blaze™ workstation and then follow the three steps described below to set up the deconvolution mode.

Step 1

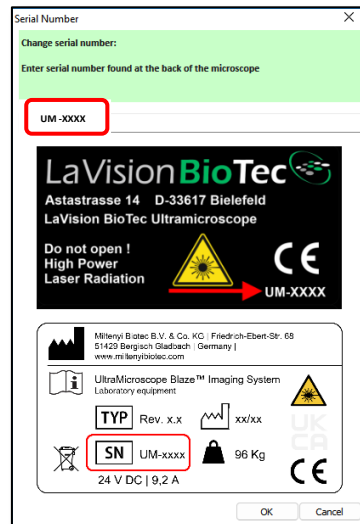
The initial step involves inputting the serial number of the Ultramicroscope Blaze into ImSpector. You can locate the serial number on the back of the instrument. Access the settings, navigate to Hardware, and select "Set Serial Number."



Installation and data acquisition

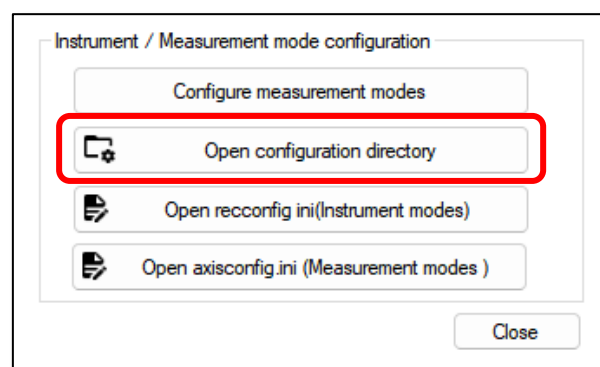
ImSpector deconvolution mode

Please input the serial number in the designated field.



Step 2

Open the "config" folder in ImSpector. To access this folder, click on  and then at the bottom of the "User Settings" menu, select "Open Configuration Directory."



After opening the configuration folder, close ImSpector and place the instrument specific calibration file (named: *umb_light_sheet_calibration.ini*) within the configuration folder. Note that the information contained in this file will be automatically written into the metadata of the acquired images.

Installation and data acquisition

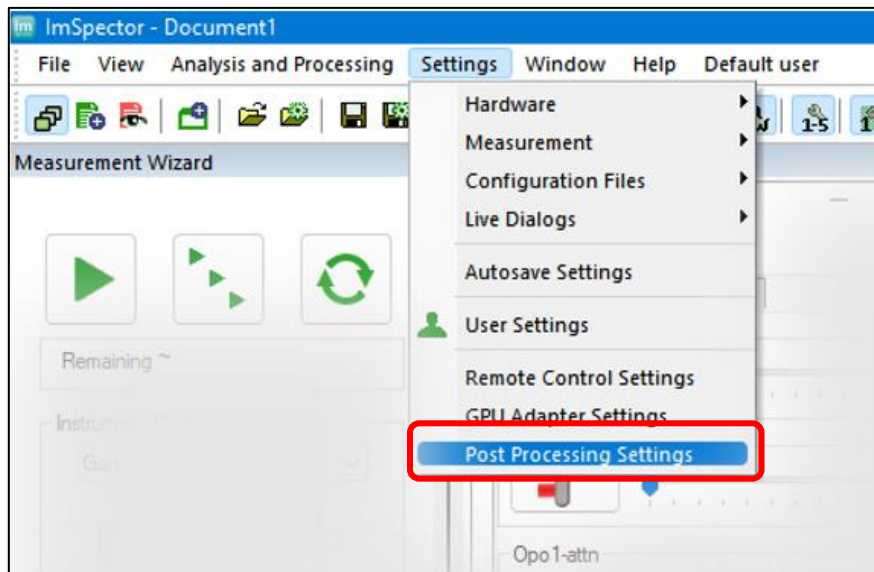
ImSpector deconvolution mode



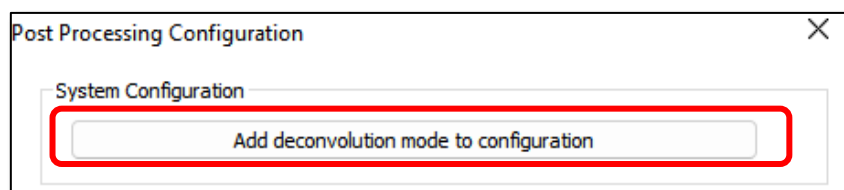
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Step 3

Now, open ImSpector again and in the "Settings" tab click on "Post Processing Settings".



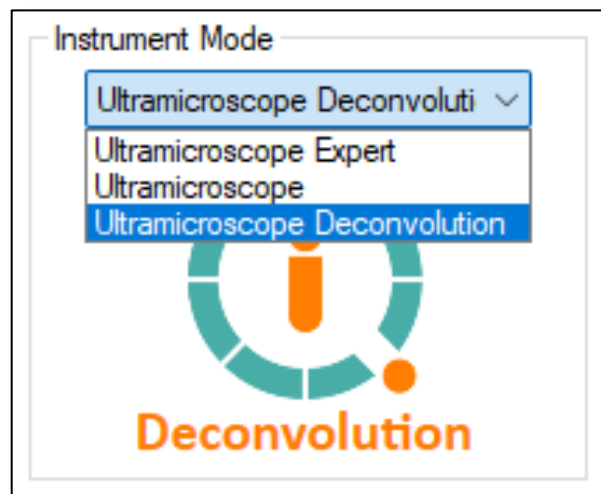
In this step, you can include the deconvolution mode by selecting "Add deconvolution mode to configuration".



After completing these steps, "Ultramicroscope Deconvolution" will be available as an option in the "Instrument Mode" tab, which will be further explained in the following chapter.

Data acquisition

In order to acquire data best suited for deconvolution, please select the “Ultramicroscope Deconvolution” from the “Instrument Mode” tab.



In this instrument mode, the provided settings are preconfigured to optimize system performance and produce the best results. While some parameters **should not or cannot be altered** to maintain essential parameter assumptions, you have the *flexibility to experiment* with others.

- **Default objective is 12×, internal magnification is set to 2.5×**

Deconvolution currently supports:

- 4× with 1.6× and 2.5× post magnification
- 12× with 1×, 1.6× and 2.5× post magnification

- **NA is set to 0.06**

An optimum between Rayleigh length and light sheet thickness, reducing the required focus positions and saving time during the analysis.

- **Only the middle light sheet per side is active**
- **Dynamic horizontal focus (DHF) is ON** with a pre-set of *three steps*, considering a range of the entire field of view (FOV)

Installation and data acquisition

ImSpector deconvolution mode



- **In the Autosave Settings – Advanced Settings, Save Raw Data is automatically activated**
- **Fixed HF positions in continuous mode**
- *Z-stacks step size is set to 1 μ m*
- *sheet width 40%*

In order to acquire a dataset, please define:

- FOV
- imaging direction (left or right)
- Z-stack step size
- Channels (with their respective power and exposure time)
- Save directory and filename
- Range and steps of the dynamic focus

Currently, only processing of images where the light sheet has been formed using a numerical aperture (NA) of 0.06 is supported. This specific NA strikes a balance between achieving a suitably narrow light sheet waist and ensuring an adequately extensive Rayleigh length.

By employing this NA configuration, it becomes feasible to minimize the required horizontal focus positions for the light sheet, based on the chosen magnification. This, in turn, decreases the time taken for imaging and the amount of data generated.

Because the light sheet needs to have an NA value of 0.06, it's important to select the necessary horizontal focal positions (HF) as effectively as possible. When using lower magnifications, the area being captured on the object's side becomes larger, requiring additional HF positions to be added. The table below displays different quantities of HF positions covering the entire Field of View (FOV), along with the resulting quality of deconvolution (ranging from 0 to ++). If unsure, it's recommended to use a minimum of three steps, as determined by the Instrument Mode.

Installation and data acquisition

ImSpector deconvolution mode



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HF positions 4× objective	2.5×	1.66×
2HF	+	+
4HF	++	++

HF positions 12× objective	2.5×	1.66×	1.0×
1HF	+	+	0
2HF	++	++	++

Current restrictions

Deconvolution is tested for the use of MACS imaging solution and ECi only. If you are utilizing different imaging solutions, we encourage you to assess your deconvolution results and share feedback with us.